Session 6 · Deck 6.B2

Transitions and the Life Cycle

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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

you are here: 6.B2 ▼

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Sep 24	Oct 1	Oct 8	Oct 15	Oct 22	Oct 29	Nov 5	Nov 12	Nov 19	Dec 3
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trunk	branches	the machine · classical methods			the wall	new tools		new data	colleagues

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Block A — Aiyagari: 6.A1 the household, from Bellman to Euler · 6.A2 the population and the price.
Block B: 6.B1 Krusell–Smith · **6.B2** transitions and the life cycle.

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Roadmap: Six Parts

One surprise, then perfect foresight: the model and its equilibria first, then the path — where it can be checked exactly, then where it cannot.

Part	Topic	The question it answers
1	The economy with a tax	The model, its stationary equilibrium, and how it is solved.
2	A reform, not a steady state	An MIT shock: what it is, why, how it is solved, what it teaches.
3	The classical model, three ways	Shooting, stacking, iterating: which survives a distribution?
4	The same shock with a distribution	The algorithm, step by step: backward, forward, compare.
5	Who gains	How the consumption-equivalent variation is built, and what it says.
6	Finite lives	The model with ten-year periods, how to compute it, what it is for.

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Idea

The method that survives a distribution — guess prices, households backward, distribution forward — is first run on the growth model, where two exact methods can check it.

The Economy: Block A's Households and Firm, Plus a Tax and a Rebate

6.A1's economy, annual, plus one policy: a tax τ on capital income, rebated to everyone as a lump sum T.

Households. A continuum, each maximizing $\mathbb{E}_0 \sum_t \beta^t u(c_t)$, $u(c) = c^{1-\sigma}/(1-\sigma)$, subject to

$$c + a' = \big(1 + (1-\tau)\, r\big)\, a + w\varepsilon + T, \qquad a' \geq 0.$$

ε follows 6.A1's Markov chain Π (mean one).

The firm. $Y = K^{\alpha} L^{1-\alpha}$ with $L = 1$: $r = \alpha K^{\alpha-1} - \delta$, $w = (1-\alpha) K^{\alpha}$.

The government. Taxes capital income at τ and rebates it all: $T = \tau r K$ to everyone.

Calibration (6.A1): $\alpha = 0.33$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$; $\rho_{\varepsilon} = 0.90$, s.d. 0.20, 7 states; 200 nodes on $[0, 150]$.

Idea

The tax touches the household in one place: the return it earns is $r^{\text{net}} = (1-\tau)r$. The rebate enters like income. Chain, limit and grid are Block A's.

The Euler equation, with the tax

$$u'(c) \;\geq\; \beta\big(1 + (1-\tau)r\big)\, \mathbb{E}\big[u'(c') \,\big|\, \varepsilon\big],$$

with equality when $a' > 0$. Only the after-tax return appears; T enters through c.

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Ledger `core.params`; `economy` in `s06_ha.py` (`r_net`, `transfer`). Block A's economy: 6.A1, "The Economy".

6.A1's definition with one more object, the rebate, and one more condition, the government's budget. Prices are numbers; the distribution sits still.

Definition

Given τ, a **stationary recursive competitive equilibrium** is prices (r, w), a rebate T, a value $v(a, \varepsilon)$, policies $c(a, \varepsilon)$, $a'(a, \varepsilon)$, a distribution λ over (a, ε) and capital K such that:

1. Households: given (r, w, T), v solves the Bellman equation with the last frame's budget, and (c, a') are its policies.

2. Firm: $r = \alpha K^{\alpha-1} - \delta$ and $w = (1-\alpha)K^{\alpha}$. **3. Labour:** $L = \int \varepsilon \, d\lambda = 1$. **4. Capital:** $K = \int a \, d\lambda$.

5. Stationarity: $\lambda(\mathcal{B}, \varepsilon') = \sum_{\varepsilon} \Pi(\varepsilon, \varepsilon') \, \lambda\big(\{a : a'(a, \varepsilon) \in \mathcal{B}\}, \varepsilon\big)$ for every set $\mathcal{B}$.

6. Government: $T = \tau \, r \, K$: what is collected is what is rebated.

Idea

At $\tau = 0$ this is 6.A1's definition word for word. The tax adds one object, T, and one equation, 6.

What the tax does to the bracket

Patience bounds the *after-tax* return, $(1-\tau)r < 1/\beta - 1$, so the pre-tax r may exceed $1/\beta - 1$: 6.A2's upper bracket becomes $(1/\beta - 1)/(1-\tau)$ (next frame).

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2.A2, "Stationary Recursive Competitive Equilibrium"; Aiyagari, *QJE* (1994); 6.A1, "The Equilibrium This Block Computes".

1 · Solving It: 6.A2's Loop, With the Tax in Two Lines

Block A's three nested problems, unchanged. Line 6 is the whole reform: the household reads $(1-\tau)r$ and T where it read r.

1: grid, ε, $\Pi \leftarrow$ power grid, Rouwenhorst ▷ 6.A1
2: $r_{\text{lo}} \leftarrow -\delta + 0.02$; $r_{\text{hi}} \leftarrow (1/\beta - 1)/(1-\tau) - 10^{-5}$ ▷ **changed**
3: evaluate $d(r_{\text{lo}})$, $d(r_{\text{hi}})$; stop unless $d(r_{\text{lo}}) < 0 < d(r_{\text{hi}})$ ▷ 6.A2
4: **while** $r_{\text{hi}} - r_{\text{lo}} > 10^{-7}$ **do**
5: $\quad r \leftarrow (r_{\text{lo}} + r_{\text{hi}})/2$; $(K^d, w) \leftarrow$ the firm at r
6: $\quad r^{\text{net}} \leftarrow (1-\tau)\, r$; $T \leftarrow \tau\, r\, K^d$ ▷ **changed**
7: $\quad c, a' \leftarrow$ household at (r^{net}, w, T), EGM, warm-started ▷ 6.A1
8: $\quad \lambda \leftarrow$ invariant distribution by iteration, warm-started ▷ 6.A2
9: $\quad d \leftarrow \sum \lambda\, a - K^d$; if $d > 0$: $r_{\text{hi}} \leftarrow r$, else $r_{\text{lo}} \leftarrow r$
10: **end while**

A tighter bracket, from the economics

The tax raises the pre-tax return, but not beyond the rate that restores the old after-tax return: there households would hold about as much as before while firms want less. So bisect on $[r_{\text{old}},\, r_{\text{old}}/(1-\tau)] = [2.356\%,\, 2.945\%]$: excess supply -1.494 and $+0.081$, a root inside, 16 bisections.

`solve_equilibrium` in `s06_ha.py`; `steady_state_after_tax` in `s06_paths.py`. **Ledger** `transition.live` (`new_ss_bracket_start_pct`, `new_ss_sign_change`, `new_ss_bisections`).

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Both Steady States, Solved

The economy before the reform ($\tau = 0$) and the one it will travel to ($\tau = 20\%$). The transition of Part 2 starts at the first and must end at the second.

	old, $\tau = 0$	**new**, $\tau = 20\%$
capital K	5.639	5.211
pre-tax r	2.356%	2.919%
after-tax return	2.356%	2.335%
wage w	1.186	1.155
rebate T	0	0.030

K falls 7.59% and the wage 2.57%. The after-tax return ends slightly *below* the old one: with precautionary saving, households keep holding capital at a lower reward.

Idea: against one household

With one household and no risk (Part 3) the after-tax return is pinned at $1/\beta - 1 = 4.167\%$ in both steady states, and the same tax lowers K by 11.5%. Here the supply of capital is less elastic, so the price absorbs part of the reform.

Ledger `transition.live` (`K_old`, `K_new`, `r_old_pct`, `r_new_pct`, `r_net_new_pct`, `w_old`, `w_new`, `rebate_new`); `ra_transition.steady_states` **for the one-household rows.**

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What an MIT Shock Is, and Why We Use It

One surprise, then perfect foresight. The name first, then why it is the right experiment for a reform — and what it buys computationally.

Three ingredients.

1. **Unanticipated.** Before $t = 0$ nobody gives it any probability: the economy rests in its old steady state.
2. **Once.** At $t = 0$ it happens — here a permanent $\tau = 20\%$ with its rebate — and never again.
3. **Perfect foresight after.** From $t = 0$ on everyone knows the whole path.

Why we use it. After the surprise nothing aggregate is random: prices are a *sequence* $\{r_t, w_t\}$, not a function of Γ — no distribution as a state, no forecast rule, none of 6.B1's loop. The unknown is one path $\{K_t\}_{t=0}^{T}$: 5.A3's impulse response, nonlinear and with a population.

Idea

Compare two steady states and you see where the economy ends up; hit it with an MIT shock and you see *how it gets there*: how fast, through which prices, who is alive meanwhile.

What it assumes

A zero-probability event, then none: households never insure against the reform; nothing after it is uncertain. For shocks that keep coming, 6.B1.

"MIT shock": the folklore name for an unanticipated, once-only, perfectly foreseen change; our reform is one. Session 5 computed the linear version (5.A3).

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How It Is Solved, and What It Teaches

Both ends of the path are known before anything is computed. In between: households backward from the end, the distribution forward from the start, and a guessed path of K corrected until the two meet.

1. **Both ends first.** λ_0 from the old steady state; v_T and the prices at T from the new one (Part 1).
2. **Guess the path** $\{K_t\}_{t=1}^{T-1}$; prices and the rebate follow, date by date.
3. **Backward.** From v_T, one Euler step per date: c_t, a'_t for $t = T-1, \ldots, 0$. No fixed point (6.A1).
4. **Forward.** From λ_0, push with a'_t: λ_{t+1} and $\hat{K}_{t+1}$. No fixed point (6.A2).
5. **Compare, damp.** $K \leftarrow (1-\eta)K + \eta\hat{K}$, K_0 fixed; repeat 2–5 until the paths agree.

What we learn

The path of K, prices and the rebate, its speed (a half-life), and v_0 for everyone alive at $t = 0$ — Part 5's input.

Example: a published transition

Social Security privatization in a life-cycle model: long-run capital $+24\%$, future newborns $+6.9\%$, **workers alive at the reform** $\mathbf{-2.9\%}$. Everyone who votes on it loses; everyone who gains is unborn.

Nishiyama & Smetters, *QJE* 122 (2007), as reported in *Handbook of Computational Economics*, vol. 3, ch. 3 (2013); transcribed.

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The Transition Equilibrium: Part 1's Definition With a Date

Part 1's conditions at every date, plus one new one: where the path starts and ends.

Definition

Given τ and the old invariant distribution λ_0, a **transition equilibrium** is prices $\{r_t, w_t\}$, rebates $\{T_t\}$, values and policies $\{v_t, a'_t\}$ and distributions $\{\lambda_t\}$, $t \geq 0$, such that:

1. **Households:** given the whole price path, v_t solves the Bellman equation with a date (Part 4); a'_t is its policy.
2. **Firm:** $r_t = \alpha K_t^{\alpha-1} - \delta$, $w_t = (1-\alpha)K_t^{\alpha}$.
3. **Government:** $T_t = \tau r_t K_t$.
4. **Capital:** $K_t = \int a \, d\lambda_t$ at every date.
5. **The distribution moves:** λ_{t+1} is λ_t pushed through a'_t and Π.
6. **Ends:** λ_0 is given, and $(r_t, w_t, v_t, \lambda_t)$ converge to the new stationary equilibrium.

Idea

Drop the dates and require $\lambda_{t+1} = \lambda_t$: this is Part 1's stationary definition. With dates, condition 5 is not a fixed point: it runs forward from a known start.

In the computation

Condition 6 is imposed at $T = 150$: from T on, everything equals the new steady state. The unknowns are $K_1, \ldots, K_{T-1}$, and condition 4 gives one equation for each.

Auerbach & Kotlikoff, *Dynamic Fiscal Policy* (1987); Part 1 for the stationary version.

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What Must Be Found: a Path, Not a Number — and the Compromise

The stationary equilibrium was one number, r. The transition is a whole path, and every household must know all of it.

Count the unknowns. Stationary: one r, found by 16 bisections (Part 1). Transition: $K_1, \ldots, K_{T-1}$ — $T-1$ numbers, one market-clearing condition each — and the household at date t needs $\{r_s, w_s, T_s\}_{s>t}$, the whole future.

The compromise. Transitions are asymptotic: the economy never quite arrives. Computational work assumes it has arrived by some T: set $v_T = v_\infty$, the new steady state's value, and solve the household *backward* from T given a price path.

Three things make this computable.

- K_0 and λ_0 are *given*: the old steady state.
- v_T is *imposed*: the new steady state.
- Perfect foresight: no expectation over prices, only over ε.

Imposed vs. produced

v_T is imposed; λ_T is *produced* by pushing λ_0 forward, and must arrive at λ_∞ on its own. If it does not, T was too short (Part 4 checks).

Idea: which method?

$T-1$ equations in $T-1$ unknowns can be shot, stacked, or iterated. Part 3 tries all three where two of them are exact — one household, no risk — and keeps the one that survives a distribution.

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Auerbach & Kotlikoff, *Dynamic Fiscal Policy* (1987), for the truncation at T.

The Classical Model With the Same Tax

Why first here: one household and no income risk, the same tax and rebate, the same calibration — and two exact methods, so Part 4's loop can be checked before a distribution is in it.

Resources (the rebate returns the tax):

$$c_t + K_{t+1} = f(K_t) \equiv K_t^{\alpha} + (1-\delta)K_t.$$

Euler equation (only the after-tax return matters):

$$c_t^{-\sigma} = \beta\big(1 + (1-\tau)\, r_{t+1}\big)\, c_{t+1}^{-\sigma}, \qquad r_{t+1} = \alpha K_{t+1}^{\alpha-1} - \delta.$$

Steady state: $(1-\tau)\, r^* = 1/\beta - 1$, so the tax raises the pre-tax return and lowers K^*.

	$\tau = 0$	$\tau = 20\%$
pre-tax r^*	4.167%	5.208%
after-tax return	4.167%	4.167%
capital K^*	4.434	3.922
consumption	1.280	1.256
wage	1.095	1.052

Capital falls 11.5% here; in Block A's economy the same tax lowers it 7.6% (Part 1).

Ledger `ra_transition.steady_states`; $\alpha = 0.33$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$, $L = 1$.

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The Equilibrium: Four Conditions, Every Date

After the surprise there is no risk, so an equilibrium is a set of sequences. The two equations on the last frame are these four conditions combined.

Definition

Given K_0 (the old steady state) and τ, a **competitive equilibrium** is sequences $\{c_t, K_{t+1}\}$, prices $\{r_t, w_t\}$ and rebates $\{T_t\}$, $t \geq 0$, such that:

1. Household: given prices and rebates, $\{c_t, k_{t+1}\}$ maximizes $\sum_t \beta^t u(c_t)$ subject to $c_t + k_{t+1} = \big(1 + (1-\tau)r_t\big)k_t + w_t + T_t$.

2. Firm: $r_t = \alpha K_t^{\alpha-1} - \delta$ and $w_t = (1-\alpha)K_t^{\alpha}$. **3. Government:** $T_t = \tau r_t K_t$. **4. Market:** $k_t = K_t$.

From the definition to the two equations

Resources: put 2–4 into the budget. Tax and rebate cancel, and $r_t K_t + w_t = K_t^{\alpha} - \delta K_t$, so $c_t + K_{t+1} = K_t^{\alpha} + (1-\delta)K_t$. **Euler:** the household's first-order condition in k_{t+1}, which sees only $(1-\tau)r_{t+1}$.

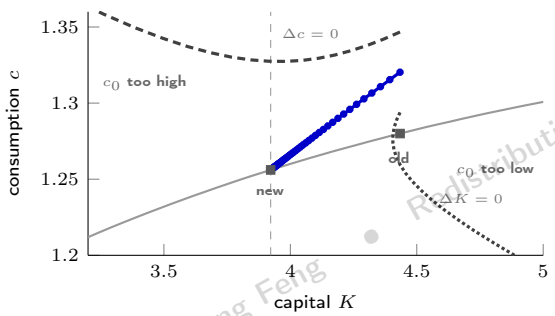
Why one more condition

Two equations in (c_t, K_t), but only one starting value, K_0. The household's transversality condition supplies the other: next frame.

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3 · A Boundary-Value Problem: Only One c_0 Works

A wrong guess does not converge: it explodes. K_0 is given, c_0 is free, and exactly one c_0 reaches the new steady state.



Once c_0 is chosen, the resource constraint and the Euler equation fix everything after it. The right $c_0 = 1.320$ (red, years 0–60) rides the saddle path to the new steady state. Start 2% above it (dashed): the household eats too much and capital collapses. Start 2% below (dotted): it saves too much and capital piles up. Both leave along the unstable root.

Idea: two roots

Linearized at the new steady state, the map $(K_t, c_t) \mapsto (K_{t+1}, c_{t+1})$ has roots $\lambda_s = 0.9225$ and $\lambda_u = 1.1405$; their product is $1 + r^* = 1.052$. One direction converges, one explodes: 5.A3's saddle, here in two variables.

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Ledger `ra_transition.fig_ra_phase` (`saddle`, `wrong_high`, `wrong_low`, `wrong_eps`), `eigen_new`, `newton.c0`.

3 · Methods 1 and 2: With One Household, Shooting and Stacking Work

The whole economy is two numbers, (K_t, c_t), so two methods work here that a distribution rules out.

Method 1: shooting

Turn the boundary-value problem into an initial-value one: guess c_0, integrate the two equations forward, read how the path fails (capital collapses: ate too much; piles up: ate too little), bisect on c_0. **Weakness:** an error in c_0 grows like $\lambda_u^t = 1.14^t$, one digit lost every 17.5 years, so long horizons are out of reach.

Method 2: stacking

Write the T Euler equations in the T unknowns $K_1, \ldots, K_T$ (K_0 given, $K_{T+1} = K^*_{\text{new}}$) as one system $F(K) = 0$. Equation t touches K_t, K_{t+1}, K_{t+2} only, so the Jacobian is tridiagonal and Newton (4.A2) solves it in 5 iterations. **Reach:** the whole path is one unknown vector — the object that sequence-space methods (Auclert et al. 2021, cited in 5.A3) linearize.

Why neither survives a distribution

Shooting needs a state you can integrate forward: a few numbers, not 1,400 masses. Stacking needs the Jacobian of those masses with respect to a 150-number path. The method that needs neither is next.

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Ledger `ra_transition.shooting`, `newton` (`iterations`), `eigen_new` (`digits_lost_per_period`).

3 · Method 3: Iterate on the Capital Path

Guess the path, compute prices, solve households backward, add up savings forward, compare, damp. No Jacobian, no unstable integration.

1: fix T; solve both steady states
2: guess $\{K_t\}_{t=0}^{T}$ with $K_0 = K_{\text{old}}$
3: **repeat**
4: r_t, w_t and rebates $\tau r_t K_t$ from K_t, every t
5: households backward from $t = T$, where they act as in the new steady state
6: capital forward from K_0: $K^s_{t+1} =$ what households save
7: gap $\leftarrow \max_t |K^s_t - K_t|/K_t$; $K \leftarrow (1-d)K + dK^s$
8: **until** gap $< 10^{-5}$

Why damp ($d < 1$)

A higher guess for K_t lowers r_t, and lower returns cut the saving meant to support it: the full step overshoots (4.A1).

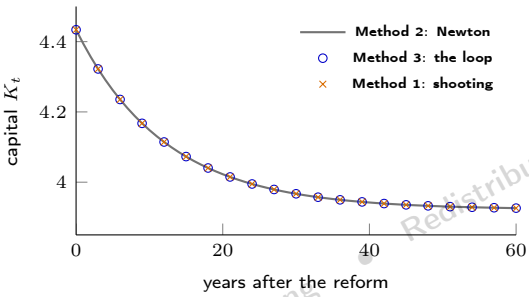
The test

Run Part 4's code unchanged with **one** household ($\varepsilon \equiv 1$) and all mass at K: with $d = 0.1$, 29 passes reproduce Newton's path to 7×10^{-6}.

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Ledger `ra_transition.degenerate_ha`; `transition_path` **in** `s06_paths.py`, **inputs from** `ra_degenerate_economy` **in** `s06_ra.py`.

All three give the same path. Only one of them still works when the household is a population.



Method 3 needs prices going backward and policies going forward — no Jacobian of a distribution, no forward integration of an unstable system. It is the one that survives 1,400 states (Part 4).

Its price: damping

Without income risk, saving reacts strongly to the path of r: at $d = 0.5$ the loop blows up by the third pass, at $d = 0.2$ by the fifth; $d = 0.1$ converges. In Part 1's economy, $d = 0.5$ works.

Ledger `ra_transition.fig_ra_K`, `degenerate_ha.other_dampings`. **Half of the gap is closed in 8.5 years (**`newton.half_life_years`**).**

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The Algorithm, Step by Step

Part 3's Method 3 with the household of Part 1 in it. Hats mark guesses; the only iteration is on the path.

0. fix T; solve both stationary equilibria (Part 1): (λ_0, v_0, K_0), $(\lambda_\infty, v_\infty, K_\infty)$
1. impose $(\lambda_T, v_T, r_T, w_T, K_T) = (\lambda_\infty, v_\infty, r_\infty, w_\infty, K_\infty)$
2. guess $\{\hat{K}_t\}$, $\hat{K}_0 = K_0$; prices $\hat{r}_t, \hat{w}_t$ from $\hat{K}_t$; rebate $\hat{T}_t = \tau \hat{r}_t \hat{K}_t$
3. households backward, $t = T-1, \ldots, 0$: $\hat{v}_t, \hat{c}_t, \hat{a}'_t$ from $\hat{v}_{t+1}$ ▷ 6.A1
4. distribution forward from λ_0: $\hat{\lambda}_{t+1} = \hat{\lambda}_t P_{\hat{a}'_t}$; $\hat{A}_t = \int a \, d\hat{\lambda}_t$ ▷ 6.A2
5. if $\max_t |\hat{A}_t - \hat{K}_t| < \text{tol}_A$ go to 6; else move $\hat{K}$ toward $\hat{A}$, back to 2
6. if $\|\hat{\lambda}_T - \lambda_T\| < \text{tol}_\lambda$ done; else increase T and start again

One loop, not three

Steps 3 and 4 are *passes*: $\hat{v}_{t+1}$ and $\hat{\lambda}_t$ are known when needed. The only loop is 2–5, on the path.

A smart first guess

Start from the shape of Part 3's one-household path instead of an exponential: 19 passes instead of 22.

`transition_path` in `s06_paths.py`; **ledger** `transition.ra_initial_guess`; **Lab 6, cell C1. Steps 3 and 4 in detail: the next two frames.**

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Households Backward: the Bellman Equation With a Date

Step 3, in detail. Given the price path, the household problem has a date. Its terminal function is known, so there is no fixed point to find — one step per period.

$$v_T = v_{\text{new}}, \qquad v_t(a,\varepsilon) = \max_{a' \geq 0} \Big\{ u\big((1+r_t^{\text{net}})a + w_t\varepsilon + T_t - a'\big) + \beta \sum_{\varepsilon'} \Pi(\varepsilon,\varepsilon')\, v_{t+1}(a',\varepsilon') \Big\}, \quad t = T-1,\ldots,0.$$

v_{new} is the new steady state's value. Both routes of 6.A1 apply: one VFI step per date, or one EGM step per date. The code takes the Euler route:

$$(c_t, a'_t) = \texttt{egm_step}\big(c_{t+1},\, r_{t+1}^{\text{net}},\, r_t^{\text{net}},\, w_t,\, T_t\big).$$

Idea

6.A1's sweep, used once per date instead of 356 times at one price: $T = 150$ steps per pass. The only change is two returns in one step — tomorrow's in the Euler equation, today's in the budget.

`backward_policies` in `s06_paths.py`; **the transfer** $T_t = \tau r_t K_t$ **is the rebate.**

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The Distribution Forward; Compare, Damp, Repeat

Steps 4 to 6, in detail. 6.A2's operator with a date: the old distribution is pushed through each period's policy, and the capital it holds is compared with the guess.

1: $\lambda_0 \leftarrow$ the old invariant distribution ▷ 6.A2
2: **for** $t = 0, \ldots, T-1$ **do**
3: $\quad \lambda_{t+1} \leftarrow$ lottery and push with a'_t ▷ 6.A2, with a date
4: $\quad K^s_{t+1} \leftarrow \sum \lambda_{t+1}\, a$
5: **end for**
6: gap $\leftarrow \max_t |K^s_t - K_t|/K_t$; $K \leftarrow 0.5K + 0.5K^s$; K_0 fixed

Did λ_T arrive?

v_T is *imposed*; λ_T is *produced* and must arrive. Here $\|\lambda_T - \lambda_{\text{new}}\|_1 = 3.5 \times 10^{-6}$. On the finer run ($n_a = 400$) moving the horizon from 300 to 450 years changes K_t by at most 5.3×10^{-7}.

Measured, $T = 150$

22 passes; the path takes 1.0 s, 2.4 s with both steady states.

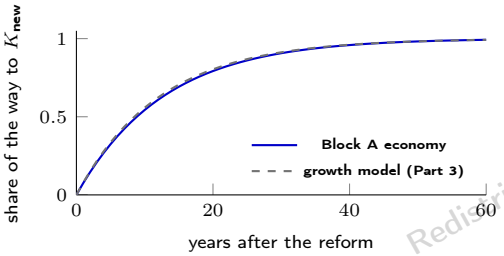
Here $d = 0.5$ converges: with precautionary saving the supply of capital is less elastic than in Part 3.

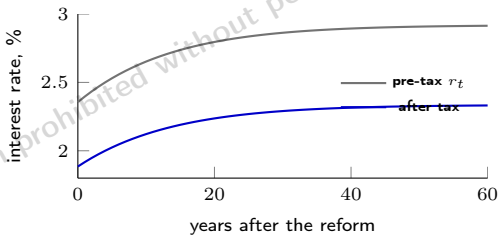
Ledger `transition.live, transition.sensitivity_T.`

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The Path, and the Checks

Capital falls 7.59%, half-way in 8.85 years; the after-tax return dips on impact and recovers.





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Checks

After tax, r falls from 2.356% to 1.885% on impact and recovers to 2.335%. Markets clear every period to 7.7×10^{-6}; goods add up ($C_t + K_{t+1} = (1-\delta)K_t + Y_t$) to 2.4×10^{-5}, a check the growth model made trivial.

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Ledger `transition.live`, `fig_adjustment` **(half-lives** 8.85 **and** 8.49 **years)**, `fig_r`.

Who Gains? The Consumption-Equivalent Variation, Step by Step

By what fraction g must the old plan's consumption rise, at every date and in every state, for a household to be indifferent to living through the reform?

1. The value of staying. In the old steady state a household at (a, ε) follows the plan $\{c_s\}_{s \geq 0}$: $v_0(a,\varepsilon) = \mathbb{E}_0 \sum_{s \geq 0} \beta^s \, c_s^{1-\sigma}/(1-\sigma)$.
2. Scale the plan by $1+g$ at every date and in every state. The factor comes out of every term:

$$\mathbb{E}_0 \sum_s \beta^s \frac{\big((1+g)c_s\big)^{1-\sigma}}{1-\sigma} = (1+g)^{1-\sigma}\, v_0(a,\varepsilon) \equiv v_0(a,\varepsilon; g).$$

3. The value of living through the reform, $v_1(a,\varepsilon)$: the date-0 value along the path, transition and new steady state included (next frame).
4. Indifference. g solves $v_0(a,\varepsilon;g) = v_1(a,\varepsilon)$, so

$$g(a,\varepsilon) = \big[v_1(a,\varepsilon)/v_0(a,\varepsilon)\big]^{\frac{1}{1-\sigma}} - 1,$$

a number in units of consumption — "0.76% more, for ever" — one per (a,ε), comparable across households and ready to add up.

Ledger `transition.cev_sanity; cev` **in** `s06_paths.py`.

Signs

With $\sigma = 2$ both values are negative. The ratio is positive, and a *less negative* v_1 gives $g > 0$.

The arithmetic, checked

Scale the old plan by 1.0123: the formula returns 0.0123 to 2.4×10^{-16}. Make the "path" the old steady state: $g = 0$ to 2.4×10^{-12}.

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5 · Where v_1 Comes From, and How g Is Added Up

v_1 is a value along a path with known policies: one backward pass, no maximization. Then average g over the households who were there at $t = 0$.

1: $V_T \leftarrow v_\infty$: the new steady state's value, from $V = u(c) + \beta\,\mathbb{E}V(a')$ at its policies
2: **for** $t = T-1, \ldots, 0$ **do**
3: $\quad V_t(a,\varepsilon) \leftarrow u\big(c_t(a,\varepsilon)\big) + \beta \sum_{\varepsilon'} \Pi(\varepsilon,\varepsilon')\, V_{t+1}\big(a'_t(a,\varepsilon),\varepsilon'\big)$ ▷ Part 4's policies
4: **end for**
5: $v_1 \leftarrow V_0$; $g(a,\varepsilon)$ from the last frame; v_0: the same recursion at the old policies, to a fixed point

Adding up. Mean of g over λ_0; the share with $g < 0$; g by wealth decile or by productivity if the question asks. Between steady states the shortcut is $g_{ss} = [v_T / v_0]^{1/(1-\sigma)} - 1$, which ignores the journey.

Idea

Line 3 is Howard's update (4.B3) with a date: the policy is given, V_{t+1} is read at a'_t by interpolation, one step per period.

Measured

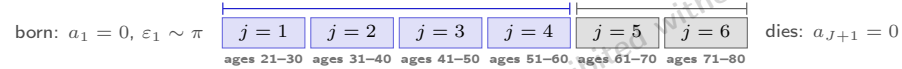
Along the path the reform gains $+0.762\%$ on average; 22.3% of households lose; the gains run from $+3.79\%$ to -2.87%.

values_along, steady_state_value, transition_welfare **in** s06_paths.py; **ledger** transition.live.welfare **(**cev_mean_pct, share_worse_off_pct, cev_max_pct, cev_min_pct**)**.

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6 · A New Model: Households That Are Born, Work, Retire and Die

Written with $J = 6$ periods of ten years: work the first $J_R = 4$, retire for two. The computed economy uses years, $J = 60$, $J_R = 45$.



The household, born with $a_1 = 0$ and $\varepsilon_1 \sim \pi$, chooses $\{c_j, a_{j+1}\}_{j=1}^{J}$ to maximize

$$\mathbb{E}\sum_{j=1}^{J} \beta^{\,j-1} u(c_j) \quad \text{s.t.} \quad c_j + a_{j+1} = (1+r)\, a_j + y_j,$$

with $a_{j+1} \geq 0$ and $a_{J+1} = 0$ (no debt; nothing left at death); ε_j follows Block A's chain Π, η_j is the age-efficiency profile (rising, then falling), u is CRRA with $\sigma = 2$.

Idea

The state is (j, a, ε); age only moves forward, and the last period is known: no fixed point to find.

Kept out, on purpose

No early death, no bequests, no labour choice; 2.B2's version has all three.

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2.B2, "**The Quantitative Life-Cycle Household**"; `olg_setup` in `s06_paths.py`.

The Rest of the Economy: Cohorts, a Firm, a Pension

One cohort is born every period and every cohort lives the same J periods, so the population is one household of each age, for ever.

Demography. A cohort of mass $1/J$ is born each period and lives exactly J periods: no population growth, no early death. At any date one cohort of each age is alive.

The firm. $Y = K^{\alpha} L^{1-\alpha}$ with r, w as in Part 1, and labour in efficiency units, $L = \frac{1}{J} \sum_{j \le J_R} \eta_j$ (ε has mean one).

The pension, pay-as-you-go. Every retiree gets $b = \theta(1-\tau_p)w$, a replacement rate θ of the average net wage. A payroll tax τ_p on workers pays for it within each period — a transfer from young to old, the government holds nothing:

$$\tau_p \, w \, \frac{J_R}{J} = b \, \frac{J - J_R}{J} \;\Rightarrow\; \tau_p = \frac{\theta \, (J - J_R)}{J_R + \theta \, (J - J_R)}.$$

Example: $\theta = 0.4$

Four working periods of six: $\tau_p = 0.4 \cdot 2/(4 + 0.8) = 16.7\%$ (arithmetic). In the computed economy, 45 working years of 60: 11.8%.

Calibration

α, β, δ, σ and the chain as in Part 1 (annual). $\eta_j = e^{0.09(j-1) - 0.0013(j-1)^2}$ in years, scaled to mean one over the working years; it peaks at age 56. $L = 0.75$.

payroll_tax, olg_at_r **in** s06_paths.py; **ledger** olg.payroll_tax_040, olg.L, olg.efficiency_peak_age.

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The Life-Cycle Equilibrium

Stationary: the cross-section by age is the same every period, though each household's assets rise and fall.

Definition

Given θ, a **stationary recursive competitive equilibrium** is prices (r, w), a pension (τ_p, b), values and policies $\{v_j, a'_j\}_{j=1}^{J}$ and distributions by age $\{\lambda_j\}_{j=1}^{J}$ such that:

1. **Households:** given (r, w, τ_p, b): $v_J(a) = u\big((1+r)a + b\big)$ and, for $j < J$, $v_j(a, \varepsilon) = \max_{a' \geq 0} \{u(c) + \beta\, \mathbb{E}\, v_{j+1}(a', \varepsilon')\}$.
2. **Firm:** $r = \alpha(K/L)^{\alpha-1} - \delta$ and $w = (1-\alpha)(K/L)^{\alpha}$.
3. **Cohorts:** λ_1 puts mass $1/J$ at $a = 0$ with $\varepsilon \sim \pi$; λ_{j+1} is λ_j pushed through a'_j and Π.
4. **Markets:** $K = \sum_j \int a \, d\lambda_j$ and $L = \sum_{j \leq J_R} \int \eta_j \varepsilon \, d\lambda_j$.
5. **Government:** $\tau_p \, wL = b \cdot \frac{J - J_R}{J}$: payroll taxes pay the pensions.

Against Aiyagari

Condition 1 starts from the known last age, condition 3 from the known first: no fixed point in either (next frame). Only r is left to find.

Condition 5, solved

With $b = \theta(1 - \tau_p)w$ and $L = J_R/J$, the budget gives τ_p in closed form (last frame).

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2.B2, "Stationary Equilibrium with Cohorts" (here without labour choice or survival risk).

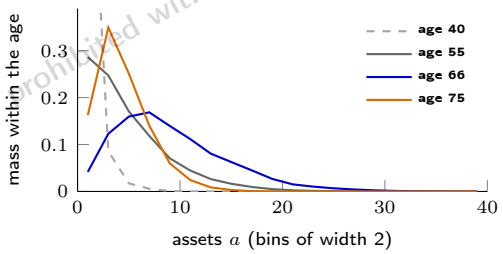
The Distribution With Cohorts: Pinned by the Newborns, Still Nested in r

In Aiyagari the distribution is a fixed point; here it is a pass over age, anchored by the newborns. The nesting in the price remains.

Aiyagari: $\lambda = \lambda P_g$, a fixed point in the space of measures (6.A2): hundreds of sweeps, or one linear solve.

Here: λ_1 is *given* — mass $1/J$ at $a = 0$, $\varepsilon \sim \pi$ — and $\lambda_{j+1} = \lambda_j P_{a'_j}$: $J - 1$ pushes of 6.A2's `push_forward`, no fixed point. The stationary cross-section is the stack $\{\lambda_j\}_{j=1}^{J}$, the same every date.

Still nested. Every a'_j depends on (r, w, τ_p, b), so the stack does: $K(r) = \sum_j \int a \, d\lambda_j(r)$. Market clearing, $K(r) = K^d(r)$, is still one equation in r, solved by the same bisection. The outer fixed point stays; only the inner one became a pass — 2.B2's "two fixed points, or one?".



At $r^* = 4.52\%$, 65% of 25-year-olds and 28% of 40-year-olds hold nothing (age 40's first bin is cut off); no 75-year-old does.

Ledger `pictures.olg_dist_by_age` ($J = 60$, $J_R = 45$, **40% replacement**). **2.B2, "Two Fixed Points, or One?".**

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6 · How to Compute It: Backward Over Age, Forward Over Age, Bisect on r

Part 1's loop with its two inner fixed points replaced by passes over age. The price loop stays.

1: bracket r on a **wide** interval; require a sign change ▷ 6.A2
2: **repeat**
3: $r \leftarrow$ midpoint; w from the firm; τ_p, b in closed form
4: **backward over age:** $c_J = (1+r)a + b$; for $j = J-1, \ldots, 1$: one EGM step ▷ 6.A1
5: **forward over age:** $\lambda_1 \leftarrow$ newborns; $\lambda_{j+1} \leftarrow \lambda_j P_{a'_j}$ ▷ 6.A2
6: $K^s \leftarrow \sum_j \int a \, d\lambda_j$; $d \leftarrow K^s - K^d(r)$; bisect
7: **until** the bracket is narrower than 10^{-9}

Why wide. With finite lives the household's patience does not cap r: the one-household bound $1/\beta - 1$ is not a bound here. Computed: $r^* = 4.52\%$, above 4.17%.

	Aiyagari	life cycle
household	fixed point (6.A1)	$J-1$ steps back
population	fixed point (6.A2)	$J-1$ pushes forward
price	bisection	bisection, wide

Computed, $J = 60$

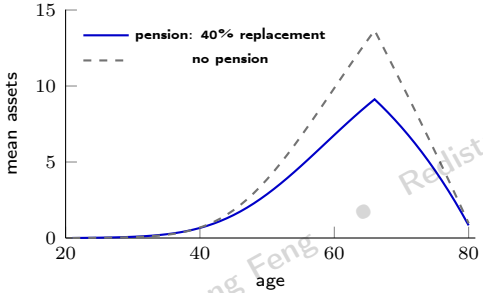
59 steps back, 59 pushes forward, 16 bisections; the pension budget balances to 4×10^{-16}.

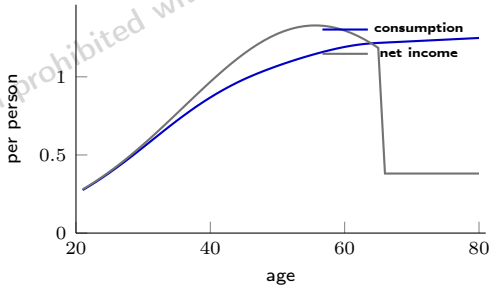
`solve_olg` in `s06_paths.py`; **ledger** `olg` (`backward_steps`, `forward_pushes`, `rows`); **2.B2 defined this economy.**

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The Life Cycle, Computed

The annual run, $J = 60$, $J_R = 45$. Households save while they work, peak at retirement and run their assets down to nothing at 80; the pension changes how much they need.



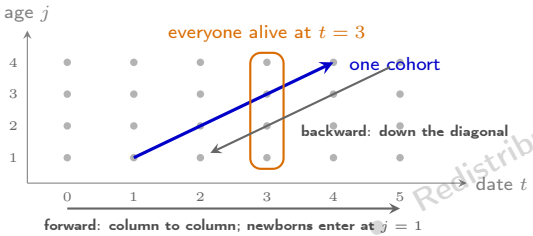


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Assets peak at 66 with or without the pension; the efficiency profile peaks at 56. Wealth Gini 0.65, above the infinitely lived economy's 0.53: age adds a dimension of inequality. **Ledger** `olg.fig_olg_assets, fig_olg_consumption, rows (peak_asset_age, gini_wealth), efficiency_peak_age.`

A Reform in This Economy: Part 4's Loop With an Age Index

Part 4's loop with an age index: backward along each cohort's diagonal, forward one date at a time. How to read the picture, then the four steps.



How to read it. A dot: one cohort at one date. Red: the same cohort ageing. Orange: everyone alive at $t = 3$, which market clearing adds up.

1. Guess a price path $\{r_t, w_t\}$ and the pension's taxes.
2. **Backward, down each diagonal**: each cohort's policies, from its last age — or from the new steady state if that lies beyond T.
3. **Forward, column to column**: the distribution over ages and assets, newborns entering at $j = 1$.
4. T clearing conditions in T unknowns: compare, damp, repeat.

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Equilibrium: the stationary definition with a date on every object, $\lambda_{j,0}$ the old cross-section, markets and the pension budget clearing every date. Auerbach & Kotlikoff (1987); Lab 6 does not run it.

Questions that have an age in them. Each needs the cohorts; most need the transition.

Question	What the model adds	A paper
Pension reform: who pays?	the cohort alive at the reform pays twice; the gain goes to the unborn (Part 2's numbers)	Auerbach & Kotlikoff (1987); Nishiyama & Smetters, *QJE* (2007)
Population ageing and r	more savers near retirement, fewer workers: the return to capital falls along the transition	Krueger & Ludwig, *JME* (2007)
The wealth distribution	age is a dimension of inequality (Gini 0.65 here vs 0.53); bequests link generations	De Nardi, *REStud* (2004)
Taxing capital or labour	the optimal mix changes when households borrow young and dissave old	Conesa, Kitao & Krueger, *AER* (2009)
Education, housing, retirement	choices with a date in a life: when to invest, buy, stop working	—

Aiyagari with a clock: cheaper to solve (no inner fixed points) and richer in what it answers, since winners and losers are cohorts.

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Citations checked against the publisher pages on 2026-10-28. Gini: ledger `olg.rows, transition.live.welfare.gini_old`.

How It Works (1): The Transition Loop and the Backward Pass

```
    for outer in range(1, max_outer + 1):
        r, r_net, w, transfer = path_prices(K, p_new, tau_k)       # prices from the guess
        c_path, ap_path = backward_policies(r_net, w, transfer, ss_new["c"],
                                            grid, e, Pi, p_new)     # households backward
        Ks, lam_T, C = forward_capital(lam0, ap_path, grid, Pi, K_old,
                                       c_path)                      # distribution forward
        resid = np.abs(Ks - K) / K
        gap = float(resid.max())
        history.append(gap)
        if verbose:
            print(f"   outer {outer:3d}   max |K^s - K| / K = {gap:.3e}")
        if gap < tol:
            break
        K = (1.0 - damping) * K + damping * Ks                      # damped update
        K[0] = K_old                                                # K_0 is given
```

```
def backward_policies(r_net, w, transfer, c_final, grid, e, Pi, p: Params):
    """Households backward from the terminal policy: one EGM step per period."""
    T = len(r_net) - 1
    c_path, ap_path = [None] * (T + 1), [None] * (T + 1)
    c_path[T] = c_final
    for t in range(T - 1, -1, -1):
        c_path[t], ap_path[t] = egm_step(c_path[t + 1], r_net[t + 1], r_net[t],
                                         w[t], transfer[t], grid, e, Pi, p)
    return c_path, ap_path
```

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How It Works (2): Forward Over Time, and Over Age

```python
def forward_capital(lam0, ap_path, grid, Pi, K0, c_path=None):
    """Distribution forward from lam0: capital supplied, lambda_T, and (given the
    consumption policies) aggregate consumption in periods 0..T-1."""
    T = len(ap_path) - 1
    Ks, C = np.empty(T + 1), np.full(T + 1, np.nan)
    Ks[0] = K0
    lam = lam0
    for t in range(T):
        if c_path is not None:
            C[t] = float((lam * c_path[t]).sum())
        j, w_hi = lottery(ap_path[t], grid)
        lam = push_forward(lam, j, w_hi, Pi)
        Ks[t + 1] = float((lam.sum(axis=1) * grid).sum())
    return Ks, lam, C
```

```python
    c_last = np.maximum((1.0 + r) * grid + b, 1e-10)     # the last year: eat everything
    c_age[I - 1] = np.repeat(c_last[:, None], ne, axis=1)
    ap_age[I - 1] = np.zeros((na, ne))
    for i in range(I - 2, -1, -1):                       # one pass over ages, no fixed point
        wage_i, pension_i = ((1.0 - tau_p) * w * eff[i], 0.0) if i < I_R else (0.0, b)
        c_age[i], ap_age[i] = egm_step(c_age[i + 1], r, r, wage_i, pension_i,
                                       grid, e, Pi, p)
        c_age[i] = np.maximum(c_age[i], 1e-10)           # no pension, no assets: eats nothing
```

```python
    lam = np.zeros((I, na, ne))
    lam[0, 0, :] = pi / I                                # newborns: no assets, e from pi
    for i in range(I - 1):                               # the cohorts ARE the distribution
        j, w_hi = lottery(ap_age[i], grid)
        lam[i + 1] = push_forward(lam[i], j, w_hi, Pi)
```

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Takeaways

- **Part 1.** The tax adds one object (T) and one equation to 6.A1's equilibrium; 6.A2's loop solves it with two lines changed and a bracket that the economics supplies.
- **Part 2.** An MIT shock is one surprise, then perfect foresight: prices become a sequence and Γ is no state. Households backward from the new steady state, the distribution forward from the old, a damped path of K between; v_T is imposed, λ_T must arrive. It teaches the path, and who lives through it.
- **Part 3.** With one household two more methods work: shooting (loses a digit every 17.5 years) and stacking (Newton, exact). The backward–forward loop matches them — with heavy damping — and is the one that survives a distribution.
- **Part 4.** With a distribution, the loop is 6.A1's step backward and 6.A2's push forward, once per date: 22 passes, K -7.59%, half-way in 8.85 years. Check λ_T and the horizon.
- **Part 5.** CEV: scale the old plan by $1+g$ until its value matches the value of living through the reform, $g = [v_1/v_0]^{1/(1-\sigma)} - 1$; v_1 is one backward pass with known policies. Mean gain 0.76%; 22% lose.
- **Part 6.** A life in J periods, one cohort per age: $J-1$ steps back, $J-1$ pushes forward from the newborns — no fixed point in either — inside a wide bisection on r (r^* may exceed $1/\beta - 1$). A reform is Part 4's loop with an age index: backward down each diagonal, forward column by column.

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Bridge: Stop Choosing the Family

Session 6 took the best classical tools to the models of Session 2 and found where each stops.

Method	What it bought	Where it stops
Grid VFI (3.A2, 6.A1)	anything, if the state is small	a grid in every state dimension
EGM (4.B3, 6.A1)	no inner search; the kink as a comparison	still a grid over the state
Perturbation (5.A)	four numbers, near a point	needs smoothness; no kinks
Chebyshev (5.B2)	spectral accuracy on a domain	needs smoothness; tensor bases
Smolyak (5.B3)	slow growth in the dimension	a box, and the work per node
Moments (6.B1)	a finite state that forecasts prices	verified at one policy only

Every row has the same shape of failure: the approximating family was **chosen in advance**, and the problem does not respect the choice.

Next: Session 7

What if the family were chosen by the problem? Machine learning for macroeconomists: networks as approximators, and the growth model solved again on its Euler residual.

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